

October

SPECIAL EVENTS

PHASES OF THE MOON

	Full Moon	New Moon
2008	October 14	October 28
2009	October 4	October 18
2010	October 23	October 7
2011	October 12	October 26
2012	October 29	October 15

THE PLANETS

2008: October 22

Mercury is at greatest morning elongation, magnitude -0.4 .

2009: October 6

Mercury is at greatest morning elongation, magnitude -0.5 .

2009: October 8

Mercury and Saturn appear less than a Moon-width apart in the eastern dawn sky.

2009: October 13

Venus and Saturn appear about a Moon-width apart in the eastern dawn sky.

2011: October 29

Jupiter is at opposition, magnitude -2.9 . At midnight, it is visible to the south from northern latitudes and to the north from southern latitudes.

2011: October 30–31

Mercury and Venus appear four Moon-widths apart in the western evening sky.

2012: October 4–5

Mercury and Saturn appear about six Moon-widths apart in the western evening sky.

2012: October 26

Mercury is at greatest evening elongation, magnitude -0.1 .

Opposition and elongation are explained on [page 14](#)

The Andromeda Galaxy, the most distant object visible to the naked eye, is on view to all observers, as is the Great Square of Pegasus, an asterism formed by linking stars in neighboring Pegasus and Andromeda. Southern observers can see both of the Milky Way's companion galaxies in the sky, and Cassiopeia is well positioned for observers in northern latitudes.

NORTHERN LATITUDES

Pegasus and Andromeda are to either side of due south. The winged horse and the princess are linked by the Great Square of Pegasus. This is formed of three stars in Pegasus and one in Andromeda, and it defines the upper body of the mythological winged horse. Close to the horse's nose is M15, a globular cluster visible to the naked eye under clear rural skies.

Andromeda is home to the Andromeda Galaxy (M31), the largest member of the Local Group cluster. It is a spiral but is tipped to our line of sight, so it looks like an elongated oval. Its central part can be seen with the naked eye and binoculars. A large telescope is needed for its spiral arms.

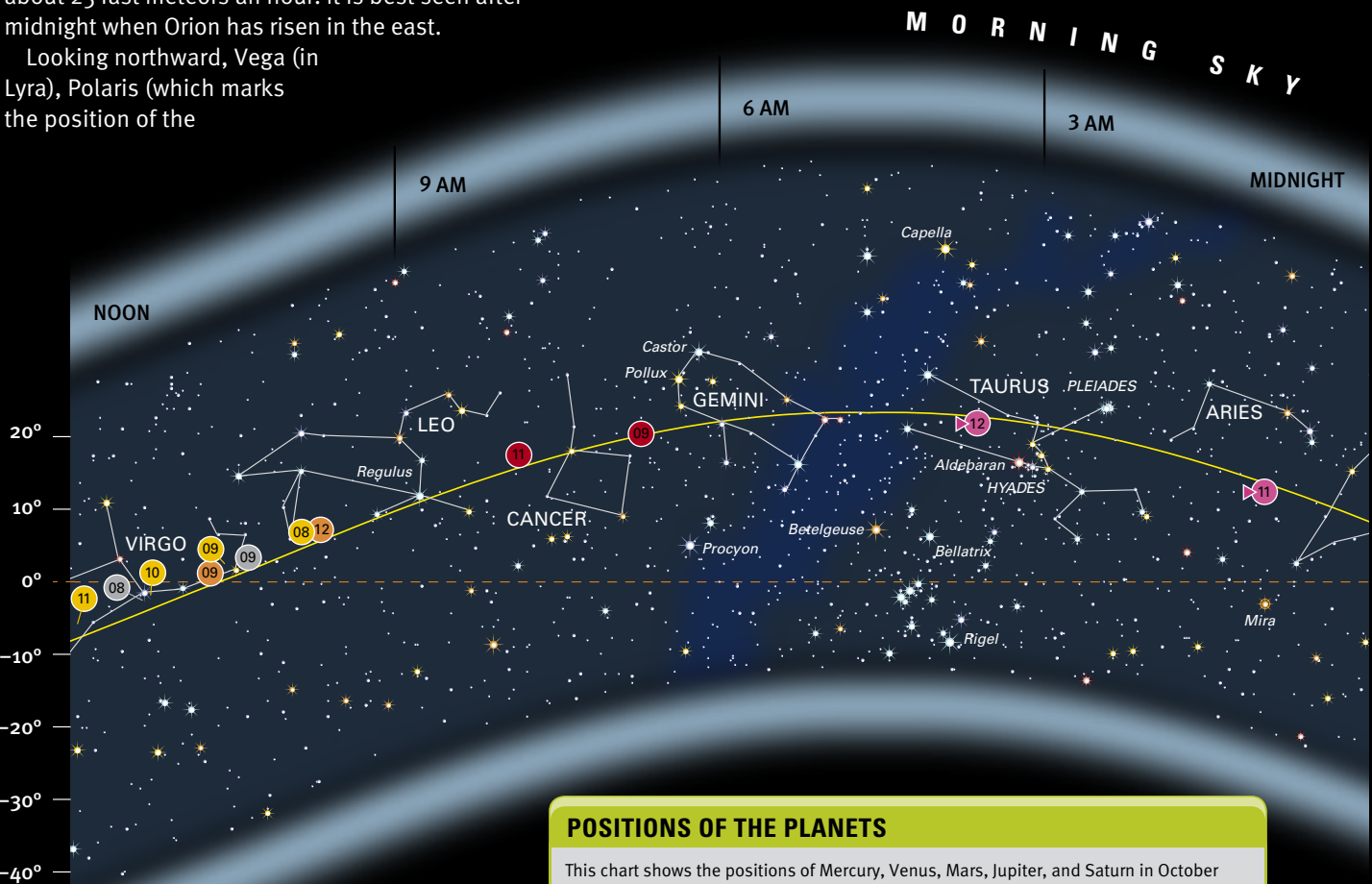
The Orionid meteor shower peaks around October 20, with about 25 fast meteors an hour. It is best seen after midnight when Orion has risen in the east.

Looking northward, Vega (in Lyra), Polaris (which marks the position of the



Cassiopeia
Here, the "M" shape of five bright stars in Cassiopeia is at upper center. Polaris is the bright star at bottom right, and two bright stars of Cepheus are to its left. The Milky Way flows through Cassiopeia from top right to bottom left.

north celestial pole), and Capella (in Auriga) make a line across the sky. The path of the Milky Way arches overhead. Cepheus and Cassiopeia are above Polaris and are in the optimum positions for viewing.



POSITIONS OF THE PLANETS

This chart shows the positions of Mercury, Venus, Mars, Jupiter, and Saturn in October from 2008 to 2012. The planets are represented by colored dots, while the number inside the dot indicates the year. For all planets apart from Mercury, the dot indicates the planet's position on October 15. Mercury is shown only when it is at greatest elongation (see p.14)—for the specific date of elongation, refer to the table to the left.

Mercury Venus Mars Jupiter Saturn

EXAMPLES

09 Mars's position on October 15, 2009

09 Saturn's position on October 15, 2009. The arrow indicates that the planet is in retrograde motion (see p.14)